

Kinetic Particle Model of Matter

Definition: Kinetic Particle Model of matter is made up of tiny particles that are in constant random motion.

#PhysicsDefinitions

Properties of solids, liquids, gasses

1. Solid

Arrangement

- Particles are closely packed and arranged in a regular pattern/orderly arrangement
- particles in solid have the least energy among 3 states of matter (kinetic energy)
- have least amount of internal energy - least potential + least kinetic

Motion

- Particles vibrate about fixed positions*
- they are held in fixed positions by very strong forces of attraction between particles.

Fixed volume and fixed shape and highest densities

2. Liquids

Arrangement

- Particles are slightly less closely packed than in solids.
- Arranged in a disorderly arrangement / Irregular pattern
- Internal energy**

Motion

- Particles can slide past/over each other throughout the liquid without fixed positions
- Forces holding the particles together are weaker than in a solid

Liquids have slightly lower densities than solids, fixed volume and no fixed shape.

3. Gas

Arrangement

- Particles are very far apart from one another in a irregular pattern/ disorderly arrangement
- Particles in gasses → have the most energy among 3 states of matter → most internal energy. Most kinetic energy + most internal energy

Motion

- Particles can move freely in any direction
- Attractive forces between particles are very weak

Gasses → Lowest densities

no fixed volume, no fixed shape.

Evidence for Movement of Particles

Brownian motion ***

How does kinetic particle Model relate to temperature

1. When a substance such as water is heated → kinetic energy of the particles increases and particle vibrate or move faster
2. When thermometer is in contact with water → more energetic particles in water transfer some of the energy to the less energetic particles in the thermometer through collisions
3. With more energy → particles tend to push each other further apart so the matter expands.
Liquids → expand more easily than solids.
Liquid column in the thermometer expands noticeably along the calibrated scale by the length of the thermometer's glass body hardly changes.**

Definition: Temperature rises with the average kinetic energy of the particles in a body and vice versa

‘average’ → implies that the concept of temperature is applicable only to a collection of particles, never to a single particle.

How does Kinetic particle model relate to pressure

Solid → only exerts pressure downwards on its base due to its weight

Fluid (liquid or gas) → exerts pressure in all directions.

When gas is trapped in a container → the particles exert pressure on all the inner walls of the container.

When a gas particle collides with the inner wall of the container → it exerts a force F_1 , on the wall.

The reaction force F_2 causes particles to bounce back from the wall.

When there are many gas particles colliding with the inner wall of the container - the forces add up.

→ the total force exerted by gas per unit area is the pressure.

Definition: Gases exert pressure, which is the average force exerted by the particles per unit area.

#PhysicsDefinitions

Gas pressure → in the container is very nearly the same throughout the container.

On the other hand → pressure due to liquid increases much faster with depth due to the much higher density of liquids than gases.**

Relationships Between Variables

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Scenario 1

Constant volume - container is sealed and volume is not allowed to change.
(Temperature changes cause pressure to change accordingly)

Relationship between temperature and pressure (T-P)

When temperature increases, Air particles average kinetic energy increases and hence move faster randomly in all directions.

Scenario 1

Air particles thus bombard against the walls of the container more vigorously (Due to increased force of each particle) and more frequently (due to increased speed of the particles)

Force of air particles hitting per unit area increases and hence the pressure within the container increases.

When temperature decreases, air particles average kinetic energy decreases and hence move slower

Air particles bombard against the walls of the container less vigorously (due to decrease force of each particle) and less frequently (due to decreased speed of the particles)

Force of air particles hitting per unit area decreases and hence the pressure within the container decreases.

Relationship:

$$Y = KX$$

NOTE: temperature and pressure graph only intersects the origin when temperature is in K(kelvin). if it is degree celsius, it will intersect at (0, -273)

Scenario 2

Constant pressure - container is allowed to expand or contract due to change in pressure in the process. (Final pressure = Initial pressure)
(temperature changes and causes volume to change accordingly)

relationship between temperature and volume (T-V)

When temperature increases, air particles average kinetic energy increases and hence they move faster.

Air particles then bombard against the inner walls of the container more vigorously (due to increased force of particles) and more frequently (due to increased speed of particles)

force of air particles hitting per unit area increases and thus the pressure within the container increases.

Pressure within the container is now greater than atmospheric pressure. there is a resultant force acting in the outward direction, which pushes the container outwards till external and internal pressure are in equilibrium.
volume increases.

When temperature decreases, air particles average kinetic energy decreases and hence they move slower.

Air particles then bombard against the inner walls of the container less vigorously

Scenario 2

(due to decreased force of the particles) and less frequently (due to decreased speed of particles).

Force of air particles hitting per unit area decreases and thus the pressure within the container decreases.

Pressure within the container is now lower than atmospheric pressure. there is a resultant force acting in the inward direction, which pushes container inwards until external and internal pressure are in equilibrium.
volume decreases.

Relationship:

$$Y = KX$$

Scenario 3

Constant temperature - change in volume of container.
can be due to expansion or contraction of volume.

change in volume causes pressure to change accordingly.

relationship between volume and pressure (V-P)

When container is being compressed, there is a larger number of air particles Per unit volume.

this results in the air particles bombarding against the inner walls of the container more frequently (due to decrease in surface area)

Force of air particles hitting per unit area increases and hence the pressure within the container increases.

When container expands, there are a smaller number of air particles per unit volume

this results in air particles bombarding against inner walls of the container less frequently (Due to decrease in surface area)

Force of air particles hitting per unit area decreases and hence pressure within container decreases.

Relationship:

$$Y = \frac{K}{X}$$

Example Questions ++

Mass of an air molecule is very small. Pressure created by the air molecules as they collide against sides of container is very large. explain:

- air molecules move about freely, and randomly in the container at high speeds. when they collide with the sides of the container, they exert a force on the container. due to the large number of air molecules colliding with sides of the container at a high rate, large force is exerted on sides of the container, which when averaged by area of the container creates a large pressure.

Suggest why Brownian motion is not observed with very large particles

- large particles of matter have more mass, and thus require a greater force, which cannot be provided by very small invisible air molecules.

Explain why pressure is exerted on the insides of the walls of a container:

- tiny particles are always in *Constant Random Motion*, the air particles collide with the inner walls of the container, generating forces per unit area of the balloon skin. thus pressure is generated.

As air is being cooled, explain what happens to internal energy:

- as air is being cooled, energy is transferred out of the air and its particles move slower. there is a decrease in energy in the kinetic store of the particles. energy in the potential store also decreases as the average separation of the particles decrease. thus internal store of air particles decreases.